

Detecting the Deceivers

An AI-Powered Approach to Fake Profile Detection

CYBERSECURITY · MACHINE LEARNING · TRUST & SAFETY



The Growing Threat

Fake profiles fuel fraud, misinformation, and manipulation at massive scale.

15%

Bot Accounts

Of all social media profiles are estimated to be fake or automated

\$78B

Annual Loss

In fraud linked to fake identities globally each year

3x

Faster Spread

Misinformation spreads 3× faster via inauthentic accounts



Traditional Detection: Where It Breaks Down



Rule-Based Filters

Easily bypassed by slightly varying patterns or behaviour

Manual Review

Too slow and unscalable for millions of new accounts daily

Keyword Matching

Misses sophisticated bots that mimic genuine language

IP Blocking

Ineffective against VPNs, proxies, and rotating addresses



Enter AI: Machine Learning Changes Everything

AI doesn't just look for red flags — it **learns** what authentic behaviour looks like, and flags what deviates from it.

Pattern Recognition

Detects subtle behavioural anomalies humans miss

Continuous Learning

Adapts as fake profiles evolve their tactics

Scale & Speed

Analyses millions of profiles in real time

Our Approach: A Multi-Faceted AI Model



No single signal is enough. Our model fuses **five data layers** to build a holistic trust score for every profile.

01

Behavioural Analysis

Posting cadence, timing, engagement patterns

02

Network Graph Analysis

Connections, clustering, and follower relationships

03

NLP Content Scanning

Language, sentiment, and copy-paste detection

04

Image Forensics

GAN-generated face detection and reverse lookup

Key Features: Suspicious Behaviour Signals



Abnormal Activity Windows

Posting at 3am daily or in inhuman bursts



Duplicate Content

Identical posts across multiple accounts simultaneously



AI-Generated Profile Photos

Detected via GAN artefact analysis



Coordinated Networks

Clusters of accounts amplifying the same content in unison



Metadata Mismatches

Location, device, and language inconsistencies

Future Forward: The Road Ahead



Multimodal Video Analysis

Extending detection to deepfake video profiles

Explainable AI (XAI)

Transparent reasoning for every flag decision

1

2

Federated Learning

Training across platforms without sharing raw data

3

4

Cross-Platform Intelligence

Unified detection across social ecosystems

Protecting Your Digital Ecosystem

Ready to build a safer, more trustworthy platform?

Project Website

 [<http://localhost:8501/>]

“This platform demonstrates our AI-based fake profile detection system designed to improve online trust and cybersecurity.”





🌟 Thank you for your time, attention, and interest in our project.

Through this AI-powered fake profile detection system, we aim to create a safer, smarter, and more trustworthy digital environment for everyone.

🔍 Together, technology and awareness can help fight online deception.